Quick Start Guide
Bruker Ultima FLIM Upgrade Kit

MU-BRK-01-1C



1 About This Guide

This guide provides the essential steps required to install and commission the FLIM LABS Bruker Ultima FLIM Upgrade Kit.

It does **not** replace the Bruker Ultima FLIM Upgrade Kit User Manual, the manuals supplied with the individual FLIM LABS components, or the documentation for any connected third-party equipment.

The complete Bruker Ultima FLIM Upgrade Kit User Manual is available at:

https://www.flimlabs.com/products/bruker-ultima-FLIM-upgrade-kit/

Scan the QR code to access the full manual.

Warning Read the full User Manual, all supplied FLIM LABS component manuals, and the documentation of any connected third-party equipment before installation and operation.

2 Safety Information

The upgrade kit integrates timing, detection, synchronization, and data acquisition hardware into an existing microscope system. Depending on the selected configuration, it may also include laser and optical components.

Safe operation requires compliance with:

- The safety instructions provided by Bruker
- The manuals supplied with the FLIM LABS hardware components
- The *FLIM IMAGER* Software Manual
- Applicable laboratory and workplace safety procedures

Warning Never look into optical fibers, microscope beam paths, or laser outputs.

Warning Enable only one excitation laser at a time unless simultaneous operation is explicitly permitted.

3 Scope of Delivery

The Bruker Ultima FLIM Upgrade Kit is supplied in a single standard configuration.

Component	Quantity
FLIM DAQ	1
Constant Fraction Discriminator Modules	2
Wideband Amplifiers	2
Active Splitter	1
Electrical connection cable set	1
USB cables and power supplies	1 set

The microscope system, including its excitation laser, photomultiplier detector and optical components, is not part of the upgrade kit.

Verify that all components of the ordered configuration are present and undamaged before installation.

4 Hardware Installation

Figure 1 shows the complete system connection diagram. If the configuration includes multiple detector channels, lasers, or synchronization paths, repeat the corresponding connections for each device or signal.

1. Install all components on stable surfaces with sufficient space for ventilation, operation, and cable routing.
2. Connect the microscope detector output to the supplied Active Splitter.
3. Connect one splitter output back to the microscope electronics and the second output to the supplied Wideband Amplifier.
4. Connect the amplifier output to the detector Constant Fraction Discriminator (CFD).
5. Connect the digital output of the detector CFD to the detector input of the *FLIM DAQ*.
6. Connect the microscope synchronization signals (pixel, line and frame) directly to the corresponding reference inputs of the *FLIM DAQ*.
7. Connect the microscope laser synchronization output to the second Wideband Amplifier, then connect its output to the second Constant Fraction Discriminator.
8. Connect the digital output of this CFD to the *SYNC-IN* input of the *FLIM DAQ*.
9. Connect the *FLIM DAQ* to the acquisition computer using the supplied USB cable.
10. Connect the required power supplies to all FLIM LABS devices.
11. Verify all optical, electrical, USB, and power connections before applying power.

5 Software Configuration

Download and install the latest version of *FLIM IMAGER* from:

https://www.flimlabs.com/products/FLIM-imager

After completing the hardware installation:

1. Power on the Bruker Ultima microscope system according to the microscope manufacturer's instructions.
2. Power on all FLIM LABS devices.
3. Verify that only one excitation laser is in the emission-active state at any time.
4. Start *FLIM IMAGER*.
5. Select *Guided Configuration*.
6. Verify that the *FLIM DAQ* is detected.
7. Detect or select the active detector channels.
8. Select the appropriate timing synchronization source.
9. Detect or configure the microscope synchronization signals.
10. Configure the image dimensions.
11. Save the configuration.

For detailed software instructions, refer to the *FLIM IMAGER* Software Manual.

6 First FLIM Measurement

1. Start the microscope scan using Prairie View.
2. Start FLIM acquisition in *FLIM IMAGER*.
3. Verify that an image appears in *Scouting Mode*.
4. Switch to *InstaFLIM* mode to acquire fluorescence lifetime images.
5. Adjust the acquisition parameters as required.
6. Save or export the acquired data.

Notice A floating acquisition control window can be opened using the W button in *FLIM IMAGER*. This allows convenient start and stop control while monitoring the microscope control software.

7 Product Label

Bruker Ultima FLIM Upgrade Kit

P/N: MU-BRK-01-1C
S/N: MU-BRK-XXXX-XXX

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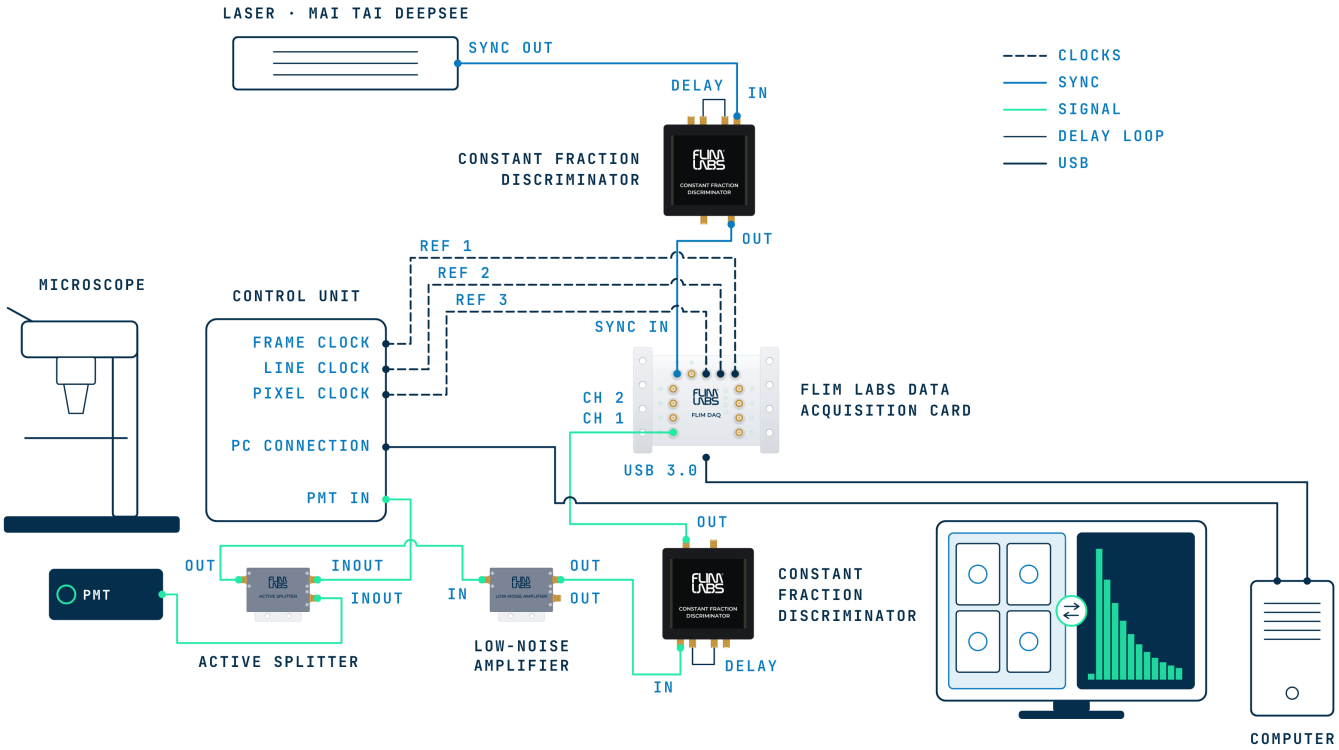


Figure 1: System connection overview.